

Project Name: TechMart Retail Management System

Document: System Analysis Document

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TechMart Retail Management System

System Analysis Document

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Customer (Entity)

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CustomerID (PK)

FirstName

LastName

CustomerPhoneNumber

CustomerEmail

CustomerAddress

Business Rule:

- A single customer can place multiple orders.

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Product (Entity)

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ProductID (PK)

ProductName

ProductDescription

ProductPrice

ProductStock

CategoryID (FK)

SupplierID (FK)

Business Rules:

- Each product belongs to one category.
- Each product is supplied by one supplier.

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Category (Entity)

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CategoryID (PK)

CategoryName

Business Rule:

- One category can contain multiple products.

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Supplier (Entity)

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SupplierID (PK)

CompanyName

PhoneNumber

Email

Address

Business Rule:

- One supplier can provide multiple products.

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Employee (Entity)

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EmployeeID (PK)

EmployeeName

JobTitle

HiringDate

Salary

BranchID (FK)

Business Rule:

- Each employee works in one branch.

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Branch (Entity)

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BranchID (PK)

BranchName

City

Address

Phone

Business Rule:

- One branch can have multiple employees.

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Order (Entity)

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OrderID (PK)

CustomerID (FK)

EmployeeID (FK)

OrderDate

OrderStatus

Business Rules:

- One customer can create multiple orders.

- One employee can handle multiple orders.

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OrderDetails (Entity)

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OrderID (FK)

ProductID (FK)

Quantity

UnitPrice

Primary Key:

(OrderID, ProductID)

Business Rules:

- One order can contain one or more products.
- One product can appear in multiple orders.
- UnitPrice stores the selling price at the time of purchase.

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Payment (Entity)

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PaymentID (PK)

OrderID (FK)

PaymentAmount

PaymentDate

PaymentMethod

Business Rules:

- One payment belongs to one order.
- Supported payment methods:
 - Cash
 - Credit Card
 - Vodafone Cash

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General Business Rules

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- Every customer has a unique CustomerID.
- Every product has a unique ProductID.
- Every supplier has a unique SupplierID.
- Every employee has a unique EmployeeID.
- Every branch has a unique BranchID.
- Every category has a unique CategoryID.
- Every order has a unique OrderID.
- Every payment has a unique PaymentID.
- A category can contain multiple products.
- A supplier can provide multiple products.
- A customer can place multiple orders.
- An order must contain at least one product.

- Product price stored in OrderDetails represents the price at the time of purchase.
- Each employee belongs to one branch.